



7.3.3 Convergence of splitting methods ¶

7.3.3 Convergence of splitting methods. Part 1

fit width

The Jacobi and Gauss-Seidel iterations can be generalized as follows. Split matrix $A = M - N$ where M is nonsingular. Now,

$$(M - N)x = y$$

is equivalent to

$$Mx = Nx + y$$

and

$$x = M^{-1}(Nx + y).$$

This is an example of a fixed-point equation: Plug x into $M^{-1}(Nx + y)$ and the result is again x . The iteration is then created by viewing the vector on the left as the next approximation to the solution given the current approximation x on the right:

$$x^{(k+1)} = M^{-1}(Nx^{(k)} + y).$$

Let $A = (D - L - U)$ where $-L$, D , and $-U$ are the strictly lower triangular, diagonal, and strictly upper triangular parts of A .

- For the Jacobi iteration, $M = D$ and $N = (L + U)$.
- For the Gauss-Seidel iteration, $M = (D - L)$ and $N = U$.

In practice, M is not inverted. Instead, the iteration is implemented as

$$Mx^{(k+1)} = Nx^{(k)} + y,$$

with which we emphasize that we solve with M rather than inverting it.

Homework 7.3.3.1. Why are the choices of M and N used by the Jacobi iteration and Gauss-Seidel iteration convenient?

► [Solution](#)

Homework 7.3.3.2. Let $A = M - N$ be a splitting of matrix A . Let $x^{(k+1)} = M^{-1}(Nx^{(k)} + y)$. Show that

$$x^{(k+1)} = x^{(k)} + M^{-1}r^{(k)}, \text{ where } r^{(k)} = y - Ax^{(k)}.$$

▼ [Solution](#)

$$\begin{aligned} &x^{(k)} + M^{-1}r^{(k)} \\ &= \\ &x^{(k)} + M^{-1}(y - Ax^{(k)}) \\ &= \\ &x^{(k)} + M^{-1}(y - (M - N)x^{(k)}) \\ &= \\ &x^{(k)} + M^{-1}y - M^{-1}(M - N)x^{(k)} \\ &= \\ &x^{(k)} + M^{-1}y - (I - M^{-1}N)x^{(k)} \\ &= \\ &M^{-1}(Nx^{(k)} + y) \end{aligned}$$

This last exercise provides an important link between iterative refinement, discussed in [Subsection 5.3.7](#), and splitting methods. Let us revisit this, using the notation from this section.

If $Ax = y$ and $x^{(k)}$ is a (current) approximation to x , then

$$r^{(k)} = y - Ax^{(k)}$$

is the (current) residual. If we solve

$$A\delta x^{(k)} = r^{(k)}$$

or, equivalently, compute

$$\delta x = A^{-1}r^{(k)}$$

then

$$x = x^{(k)} + \delta x$$

is the solution to $Ax = y$. Now, if we merely compute an approximation,

$$\delta x^{(k)} \approx A^{-1}r^{(k)},$$

then

$$x^{(k+1)} = x^{(k)} + \delta x^{(k)}$$

is merely a (hopefully better) approximation to x . If $M \approx A$ then

$$\delta x^{(k)} = M^{-1}r^{(k)} \approx A^{-1}r^{(k)}.$$

So, the better M approximates A , the faster we can expect $x^{(k)}$ to converge to x .

With this in mind, we notice that if $A = D - L - U$, where D , $-L$, and $-U$ equals its diagonal, strictly lower triangular, and strictly upper triangular part, and we split $A = M - N$, then $M = D - L$ is a better approximation to matrix A than is $M = D$.

Ponder This 7.3.3.3. Given these insights, why might the symmetric Gauss-Seidel method discussed in [Homework 7.3.2.4](#) have benefits over the regular Gauss-Seidel method?

Loosely speaking, a sequence of numbers, $\chi^{(k)}$ is said to converge to the number χ if $|\chi^{(k)} - \chi|$ eventually becomes arbitrarily close to zero. This is written as

$$\lim_{k \rightarrow \infty} \chi^{(k)} = \chi.$$

A sequence of vectors, $x^{(k)}$, converges to the vector x if for some norm $\|\cdot\|$

$$\lim_{k \rightarrow \infty} \|x^{(k)} - x\| = 0.$$

Because of the equivalence of norms, if the sequence converges in one norm, it converges in all norms. In particular, it means it converges in the ∞ -norm, which means that $\max_i |\chi_i^{(k)} - \chi_i|$ converges to zero, and hence for all entries $|\chi_i^{(k)} - \chi_i|$ eventually becomes arbitrarily small. Finally, a sequence of matrices, $A^{(k)}$, converges to the matrix A if for some norm $\|\cdot\|$

$$\lim_{k \rightarrow \infty} \|A^{(k)} - A\| = 0.$$

Again, if it converges for one norm, it converges for all norms and the individual elements of $A^{(k)}$ converge to the corresponding elements of A .

Let's now look at the convergence of splitting methods. If x solves $Ax = y$ and $x^{(k)}$ is the sequence of vectors generated starting with $x^{(0)}$, then

$$\begin{aligned} Mx &= Nx + y \\ Mx^{(k+1)} &= Nx^{(k)} + y \end{aligned}$$

so that

$$M(x^{(k+1)} - x) = N(x^{(k)} - x)$$

or, equivalently,

$$x^{(k+1)} - x = (M^{-1}N)(x^{(k)} - x).$$

This, in turn, means that

$$x^{(k+1)} - x = (M^{-1}N)^{k+1}(x^{(0)} - x).$$

If $\|\cdot\|$ is a vector norm and its induced matrix norm, then

$$\|x^{(k+1)} - x\| = \|(M^{-1}N)^{k+1}(x^{(0)} - x)\| \leq \|M^{-1}N\|^{k+1}\|x^{(0)} - x\|.$$

Hence, if $\|M^{-1}N\| < 1$ in that norm, then $\lim_{i \rightarrow \infty} \|M^{-1}N\|^i = 0$ and hence $x^{(k)}$ converges to x . We summarize this in the following theorem:

Theorem 7.3.3.1. Let $A \in \mathbb{R}^{n \times n}$ be nonsingular and $x, y \in \mathbb{R}^n$ so that $Ax = y$. Let $A = M - N$ be a splitting of A , $x^{(0)}$ be given (an initial guess), and $x^{(k+1)} = M^{-1}(Nx^{(k)} + y)$. If $\|M^{-1}N\| < 1$ for some matrix norm induced by the $\|\cdot\|$ vector norm, then $x^{(k)}$ will converge to the solution x .

Because of the equivalence of matrix norms, if we can find *any* matrix norm $\|\cdot\|$ such that $\|M^{-1}N\| < 1$, the sequence of vectors converges.

Ponder This 7.3.3.4. Contemplate the finer points of the last argument about the convergence of $(M^{-1}N)^i$

7.3.3 Convergence of splitting methods, Part 2

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Understanding the following observation will have to wait until after we cover eigenvalues and eigenvectors, later in the course. For splitting methods, it is the spectral radius of a matrix (the magnitude of the eigenvalue with largest magnitude), $\rho(B)$, that often gives us insight into whether the method converges. This, once again, requires us to use a result from a future week in this course: It can be shown that for all $B \in \mathbb{R}^{m \times m}$ and $\epsilon > 0$ there exists a norm $\|\cdot\|_{B,\epsilon}$ such that $\|B\|_{B,\epsilon} \leq \rho(B) + \epsilon$. What this means is that if we can show that $\rho(M^{-1}N) < 1$, then the splitting method converges for the given matrix A .

Homework 7.3.3.5. Given nonsingular $A \in \mathbb{R}^{n \times n}$, what splitting $A = M - N$ will give the fastest convergence to the solution of $Ax = y$?

▼ [Solution](#)

$M = A$ and $N = 0$. Then, regardless of the initial vector $x^{(0)}$,

$$x^{(1)} := M^{-1}(Nx^{(0)} + y) = A^{-1}(0x^{(0)} + y) = A^{-1}y.$$

Thus, convergence occurs after a single iteration.

7.3.3 Convergence of splitting methods, Part 3

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